



Politecnico
di Torino

Generative Artificial Intelligence for graphics and multimedia

Course introduction
Prof. Lia Morra



Teaching staff



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Organization

In-class lessons 60h + 20h laboratory

- Fri. 16:00 – 19:00, Room 2I

Class activities

- Lectures + interactive activities
- Theory-oriented (roughly 60%) and application-oriented (40%)

Lab activities:

- Toy problems to understand the basic principles
- Open-source platforms (ComfyUI/MLEagents) to build applications

Learning outcomes

Understand the basic principles behind different families of generative models (autoregressive models, variational autoencoders, adversarial networks, normalizing flows, and latent diffusion models) and **compare** their properties.

Understand the basic principles of reinforcement learning, its applications in generative AI.

Know the applications of generative AI in multimedia, animation, and computer graphics, and **design** its integration within creative pipelines.

Use state-of-the-art tools and frameworks to **train** and fine-tune generative models for conditional and unconditional synthesis of images and graphics assets.



Visualization by Gemini
(Nano Banana Pro)

Learning outcomes



Understand the basic principles behind different families of generative models (autoregressive models, variational autoencoders, adversarial networks, normalizing flows, and latent diffusion models) and **compare** their properties.

Generative models

Core idea

Observed data — images, text, video, graphs — are treated as samples from an underlying probability distribution p_{data}

A generative model learns a parametric approximation $p_{\theta} \approx p_{data}$

Formal Goal:

$$\min_{\theta} \in M \ d(p_{data}, p_{\theta})$$



Source Gemini

Learning a generative model

Parametric Models

All dataset information is compressed into a finite set of parameters θ .

- Scale with large datasets
- Trade-off: limited distribution family

The Challenge

A 12MP image has $256^3 \times 10^7$ possible values.
ImageNet has only $\sim 15\text{M}$ samples.

$256^3 \times 10^7$

possible images from a modern phone camera

15M

ImageNet images

→ **Massively underdetermined — structure discovery is key.**

Major generative model families

Autoregressive Models

GPT, PixelCNN, WaveNet

Exact likelihood

Variational Autoencoders

VAE, β -VAE, VQVAE

Latent space

Generative Adversarial Nets

GAN, StyleGAN, BigGAN

Photo-realism

Normalizing Flows

Glow, RealNVP, MAF

Exact inference

Diffusion Models

DDPM, Stable Diffusion, DALL-E

State of the art

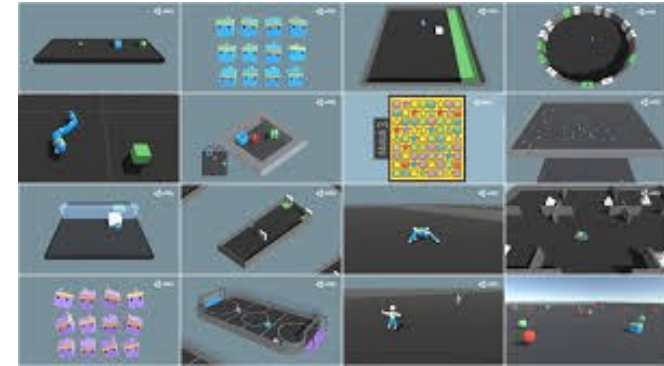
Energy-Based Models

EBM, NCSN, ScoreNet

Flexible density

Learning outcomes

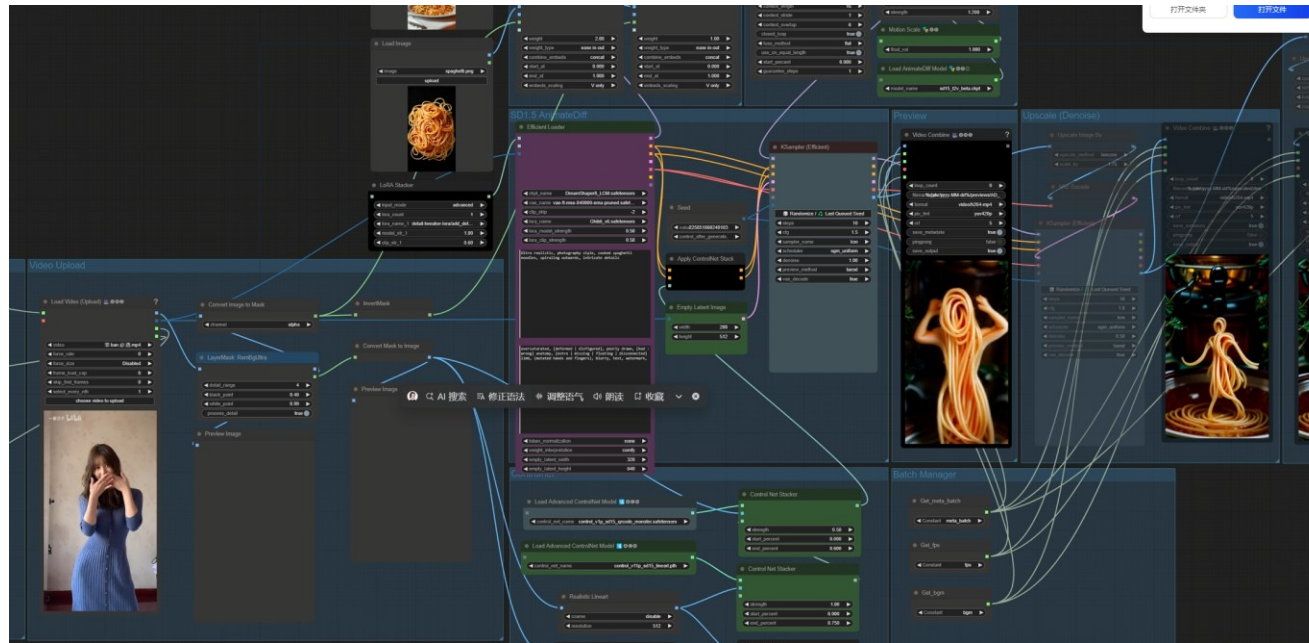
Understand the basic principles of reinforcement learning, its applications in generative AI.



Applications

Know the applications of generative AI in multimedia, animation, and computer graphics, and **design** its integration within creative pipelines.

Use state-of-the-art tools and frameworks to **train** and fine-tune generative models for conditional and unconditional synthesis of images and graphics assets.



Source <https://comfyui.org/en/video-to-animation-with-ai>

Exam

- **Group project** (25/30) and an **oral** exam (5/30). The evaluation of the oral exam is individual.
- Group project guidelines:
 - Students will work in groups (2-3 students) based on a **self-selected** problem.
 - **Outcome**: 8-page PDF report in IEEE paper format + code/resources.
 - Deadlines project proposal: 20/04/2026
 - Final submission: 7 days before the oral exam, handled via Moodle
 - Up to +2 extra points will be awarded to projects that propose original extensions, tackle particularly challenging problem settings, or include well-justified and carefully executed experiments.
- Computational resources (Crownlabs) will be provided **during** the course.
- Oral Presentation and Theory questions: 20 minute presentation, followed by QA and one theory question. Expected equal contribution of all the authors to the project work. The content and timing of the presentation will be also evaluated.
- Honors (lode) for grade ≥ 32 .

Homeworks

- Up to +3 points will be awarded for homeworks related to lab.
- Submission via Moodle.
- Collaboration policy:
 - You can show or discuss the homework solution with fellow students
 - Each student must submit **original** work

Homework	Published	Due
HW1 – GAN	03/03/2026	16/03/2026
HW2 – DiffModel	24/03/2026	16/04/2026
HW3 – MLAgents	28/04/2026	07/05/2026